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BRIEF REPORT

Dimensionality Reduction for Countermovement Jump Metrics

Lachlan P. James, Haresh Suppiah, Michael R. McGuigan, and David L. Carey

Purpose: Dozens of variables can be derived from the countermovement jump (CMJ). However, this does not guarantee an increase in useful information because many of the variables are highly correlated. Furthermore, practitioners should seek to find the simplest solution to performance testing and reporting challenges. The purpose of this investigation was to show how to apply dimensionality reduction to CMJ data with a view to offer practitioners solutions to aid applications in high-performance settings. ***Methods:*** The data were collected from 3 cohorts using 3 different devices. Dimensionality reduction was undertaken on the extracted variables by way of principal component analysis and maximum likelihood factor analysis. ***Results:*** Over 90% of the variance in each CMJ data set could be explained in 3 or 4 principal components. Similarly, 2 to 3 factors could successfully explain the CMJ. ***Conclusions:*** The application of dimensional reduction through principal component analysis and factor analysis allowed for the identification of key variables that strongly contributed to distinct aspects of jump performance. Practitioners and scientists can consider the information derived from these procedures in several ways to streamline the transfer of CMJ test information.

Keywords: performance assessment, power, athlete monitoring, analysis

Countermovement jump (CMJ) assessment is a commonly used test in strength and conditioning to inform the training process. For example, it can be used to explore neuromuscular function, assess the response to training interventions, monitor athletes' performance readiness through a competitive season, distinguish elite performers in a sport, and assess fatigue status.[1,2] The appeal of the CMJ to practitioners can be attributed to the minimal familiarization required from both tester and performer, its quick administration, the limited resultant fatigue, and the ability to yield multiple performance variables of interest.[3] Indeed, while jump height represents the outcome of the action, the analyst is often interested in the sources of the outcome to acquire a more thorough understanding of the explosive qualities of the leg extensors.

The rise in accessibility of instrumentation to quantify CMJ performance has provided practitioners with the opportunity to collect high volumes of data swiftly and more affordably than previously. However, this can present challenges, with dozens of CMJ variables reported in the literature,[4,5] and common CMJ assessment systems allow the user the freedom to consider an extensive spectrum of variables acquired from a single CMJ action. Increasing the number of variables outputted from a CMJ action does not guarantee an increase in useful information because many of the variables recorded may be highly correlated (ie, contain the same information) within a given jump. It is also important to consider that individuals are limited in their immediate capacity to process pieces of information and to make informed judgments based upon this. Therefore, practitioners should seek to find the simplest solution to a proposed question.[6] As such, achieving parsimony in CMJ metrics will aid in the interpretation of the testing.

James, Suppiah, and Carey are with the Dept of Dietetics, Nutrition and Sport, School of Allied Health, Human Services and Sport, La Trobe University, Melbourne, VIC, Australia. McGuigan is with the School of Sport and Recreation, Auckland University of Technology, Auckland, New Zealand. James (l.james@latrobe.edu.au) is corresponding author.

Principal component analysis (PCA) is a technique that reduces the dimensionality of a data set by computing a new set of variables (called principal components [PCs]). The PCs are uncorrelated and ordered so that each PC captures as much of the variation in the original variables as possible.[7] An alternative approach is a factor analysis (FA) that assumes the data can be expressed as linear combinations of a fewer number of underlying factors.[7]

This investigation shows how to apply dimensionality reduction (PCA and FA) to 3 sets of CMJ data from different devices with a view to providing practitioners with solutions to aid application in high-performance settings.

Methods

Data Collection

The CMJ data were collected from 3 cohorts (set 1: recreationally trained males, n = 16, age: 25.5 [4.2] y, body mass: 79.8 [11] kg; set 2: competitive mixed martial arts athletes, n = 29, age: 27.6 [5.5] y, body mass: 81.6 [11.6] kg; and set 3: junior elite Australian football players, n = 19, age: 17.8 [0.9] y, body mass: 77.4 [5.4] kg) across 3 separate devices (set 1: force platform [Bertec Corporation, Columbus, OH] sampling at 2000 Hz, with data processed using a custom MATLAB script [The MathWorks, Inc, Natick, MA]; set 2: linear position transducer [MuscleLab, Ergotest Technology, Langesund, Norway] sampling at 100 Hz, with data processed using a custom MATLAB script [The MathWorks, Inc]; and set 3: force platform [ForceDecks Model FD4000; NMP Technologies Ltd, London, United Kingdom] sampling at 1000 Hz, with data processed using native software [ForceDecks, version 1.2.6780; NMP Technologies Ltd]). Three trials were performed in each of the sets, and the jump with the greatest peak velocity (set 1), peak power (set 2), or jump height (set 3) was brought forward for analysis. The cue given to the participants across all sets was to "jump for maximal height." The variables collected in each data set are listed in Table 1. These variables were

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Table 1 Characteristics of Each Countermovement Jump Data Set

	Set 1	Set 2	Set 3
Variables recorded	Average power Average velocity Peak force Peak velocity Peak power Peak displacement Force at peak power Velocity at peak power Rate of power development Rate of force development Impulse	Average rate of power development Average power Average velocity Peak power Peak force Peak velocity Force at peak power Velocity at peak power Time to peak force Time to peak power Time to peak velocity	Concentric impulse Concentric mean power/BM Concentric peak force/BM Concentric peak velocity Concentric RPD Eccentric braking impulse Eccentric braking RFD Eccentric peak velocity Flight time:contraction time Force at peak power Jump height Movement start to peak force Movement start to peak power Peak power/BM Total work Velocity at peak power
Bartlett's test of sphericity	$\chi^2 = 310.9$, df = 55, $P < 2.22e{-}16$	$\chi^2 = 456.1$, df = 55, $P < 2.22e{-}16$	$\chi^2 = 608.3$, df = 120, $P < 2.22e{-}16$
Kaiser–Meyer–Olkin measure	0.62	0.65	0.51

Abbreviations: BM, body mass; RFD, rate of force development; RPD, rate of power development.

chosen as they represent metrics commonly extracted from the CMJ.

Statistical Analysis

The factorability of each data set was tested using Bartlett's test of sphericity and the Kaiser–Meyer–Olkin measure.[8,9] These assess whether correlations exist between the variables in the data (Bartlett's test)[8] and the strength of the partial correlations in the data relative to the full correlations (Kaiser–Meyer–Olkin measure).[9]

The PCA was performed on each data set using the singular value decomposition method after all variables were centered and scaled.[7] The results of the PCA were reported as the variance explained by each PC and correlations between each variable and the PCs.

The relationship between each variable and the possible underlying common factors was estimated using a maximum likelihood FA. The number of factors to extract was determined by visualizing the eigenvalues of each factor with a scree plot. Varimax rotation was applied, and the results were reported as the loadings of each original variable on the estimated underlying factors.

All statistical analyses were performed using the R statistical programming language.[10] PC and FA were implemented using the *FactoMineR*[11] and *psych*[12] packages.

Results

Table 1 shows the factorability characteristics of each data set. As expected, all data sets had significant amounts of between-variable correlation (Bartlett's test). All data sets also had an acceptable Kaiser–Meyer–Olkin measure.[9]

Principal Component Analysis

Figure 1A–1C shows the percentage of variance explained by each PC in the 3 data sets. About 90% of the total variance in the data was explained by 3 components for data set 1 and by 4 components for sets 2 and 3. The correlations between the original variables and the PCs are shown in Figure 1D–1F.

Factor Analysis

Plotting the eigenvalues of each factor (Figure 2A–2C) suggested that 2 factors should be retained for data sets 1 and 2, and potentially, 3 to 4 factors should be retained for data set 3. Figure 2D–2F shows the factor loadings of each estimated factor on the original variables.

Discussion

The primary aim of this report was to demonstrate how a high volume of common variables extracted from the CMJ test could be reduced to aid interpretability while retaining as much of the original information as possible. The application of dimensional reduction through PCA and FA allowed for the identification of key variables that strongly contributed to distinct aspects of jump performance, while the application of FA also highlights underlying, potentially trainable constructs within the CMJ. Practitioners and scientists can consider the information derived from these procedures in several ways to streamline the transfer of CMJ test information.

The application of PCA to the data sets in this study found 90% or more of the total variance could be captured by 3 or 4 PCs (Figure 1). Thus, a sports scientist seeking an efficient representation of the data may use the PCs directly when undertaking research. For example, the response to a training intervention could be quantified by changes in only 2 or 3 PCs,[13] and the researcher could be confident in the explanatory power of those few components. However, because the components are weighed sums of the original variables, they may be challenging to interpret by a practitioner, coach, or athlete.

If interpretability of monitoring variables is essential, the sports scientist could examine the correlations of the original variables with the PCs or factors and select one variable that correlates strongly with each component.[14] For example, given the results of the PCA on data set 1, the following choices of variables to retain could be made from PC1, PC2, and PC3, respectively: average power, average velocity, and peak displacement. This process also enables the practitioner some preference in the variables they select, as there

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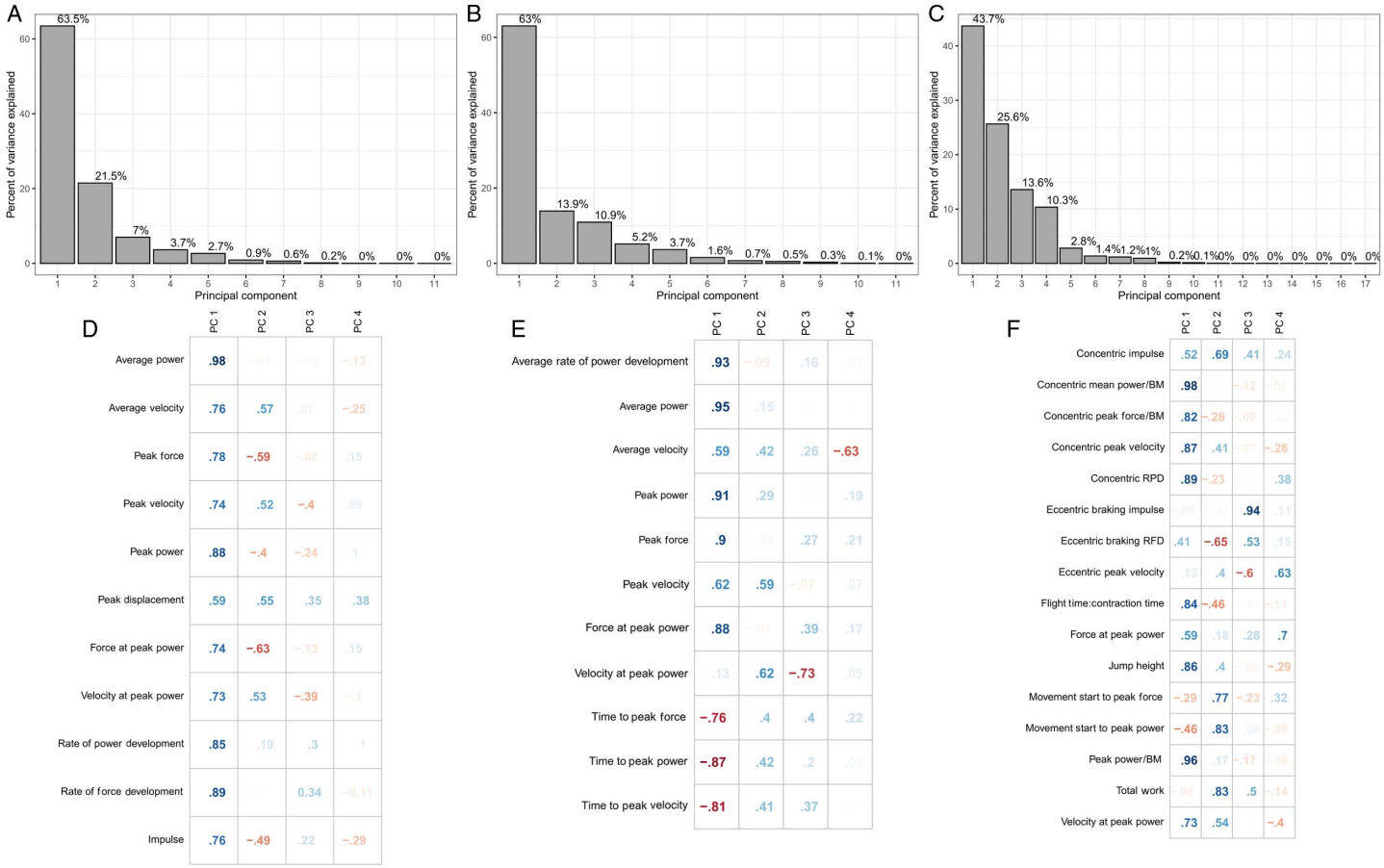

Figure 1 — PCA results: (A–C) variance explained by each PC for data sets 1, 2, and 3 and (D–F) correlation between each of the original CMJ variables and the first 4 PCs for data sets 1, 2, and 3. BM indicates body mass; CMJ, countermovement jump; PC, principal component; PCA, principal component analysis; RFD, rate of force development; RPD, rate of power development.

are often multiple variables that are highly correlated to each component. For example, from the PCA on data set 1, the following variables may be selected from components 1, 2, and 3, respectively: peak power, peak velocity, and rate of force development. In such cases where several options are available, the practitioner may choose to select a metric with the highest reliability. Although this approach will capture less of the variance than using an intact component, it is likely a more suitable option in applied settings.[6]

Similarly, from the FA, 1 or 2 newly constructed factors could be chosen to represent jump characteristics, and they could be monitored throughout the training process. An example of this, using the FA on data set 2, would be retaining "time to peak velocity" to represent factor 1 and "peak force" for factor 2 to represent different qualities of the jump. Additionally, the underlying constructs elucidated by the FA may guide training towards distinct neuromuscular capacities underpinning CMJ performance. For example, the FA appears to generate factors defined by either output-type metrics (ie, velocities and displacements), causal (ie, force and impulses), or timing (ie, time to peak velocity/power/force) and, therefore, may enable the practitioner to detect changes in underlying jump strategy (such as an altered movement time) that may not necessarily be revealed if only displacement was examined.[15]

Although the statistical approaches used in this investigation are common in research, practitioners in the field may need to invest time to familiarize themselves with these procedures. However, the increase in decision-making efficiency resulting from considering less data will certainly outweigh initial time investments. Practitioners must also apply an informed needs analysis to their sport to ensure that any variable selected is indeed relevant (ie, holds construct validity) to performance in the sport of interest. Finally, as the contents of components and factors are likely influenced by many factors (ie, population, time of season), it should not be assumed that analyses outputs are transferrable across samples. Nonetheless, this article presents a highly applicable methodological approach to CMJ data analyses.

Practical Applications

The PCA and FA can be considered by applied sports scientists to objectively reduce the number of variables needed to describe performance in the CMJ. This simplifies the integration of the test into the training environment by focusing only on test variables that contain different information.

Conclusions

The methods outlined in this study provide practitioners with a quantitative approach to determine and justify the use of a reduced

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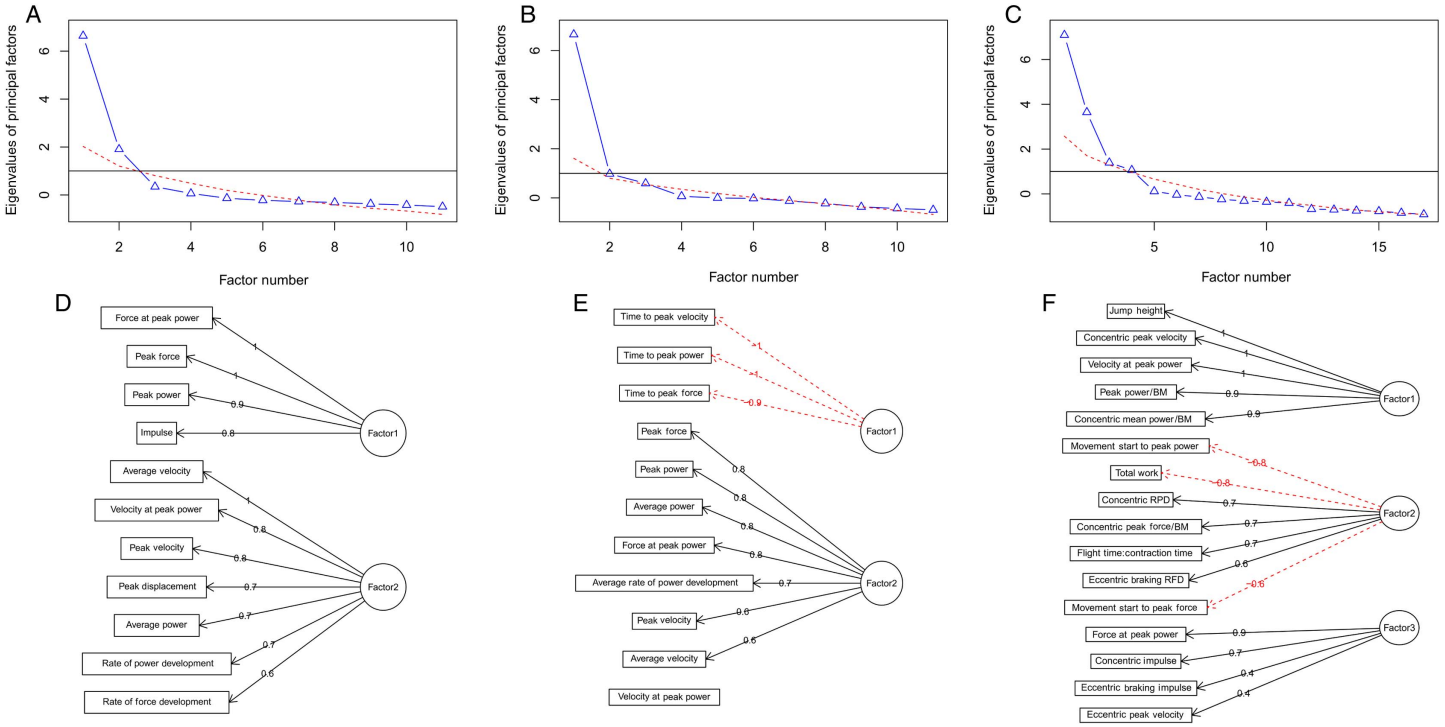

Figure 2 — Factor analysis results: (A–C) scree plots for data sets 1, 2, and 3, showing the eigenvalues of each latent factor, and (D–F) diagrams showing the standardized factor loadings between the extracted factors and CMJ variables for data sets 1, 2, and 3. BM indicates body mass; CMJ, countermovement jump; RFD, rate of force development; RPD, rate of power development.

number of CMJ performance variables for the monitoring of performance and neuromuscular fatigue. While a multitude of variables can be extracted from the CMJ, the findings from this investigation highlight that the testing procedure may not necessarily benefit from analyzing an overly extensive array of variables.

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